

04 — Management Plan

TriadicFrameworks: An Open-Science Substrate for Regime-Aware Modeling, Scanning, and Simulation

Submitted to NASA High Priority Open-Source Science (HPOSS)

1. Project Leadership and Responsibilities

The TriadicFrameworks Research Initiative will be led by:

Principal Investigator (PI):

Nawder Loswin

Founder, TriadicFrameworks Research Initiative

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The PI is responsible for:

- overall project direction and scientific integrity,
- coordination of protocol development (dsrsp/0.1, RSM, vST, SLRP),
- oversight of open-source releases and documentation,
- milestone tracking and reporting,
- community engagement and dissemination.

The PI will devote effort to technical development, documentation, and project coordination throughout the 12-month performance period.

2. Project Structure and Workflow

The project is organized into four coordinated workstreams:

Workstream A — Protocol Development

- finalize dsrsp/0.1 specification
- complete RSM and vST alignment layers
- produce schemas and reference examples

Workstream B — Reference Implementations

- develop open-source libraries for dsrsp/0.1
- implement RSM structural envelope generation
- implement vST validation block generation
- integrate ILP module

Workstream C — Integration and Testing

- integrate dsrsp/0.1 into simulation environments
- validate regime classification workflows
- generate example datasets and reproducible notebooks

Workstream D — Documentation and Community Release

- produce developer documentation
- publish integration guides
- prepare training materials
- host a virtual community workshop

Each workstream is designed to be modular, allowing parallel development while maintaining clear interfaces and deliverables.

3. Collaboration and External Engagement

TriadicFrameworks is an open-science initiative, and collaboration is central to its mission. The project will:

- maintain public repositories for code, specifications, and documentation,
- invite community contributions through issues, pull requests, and discussions,
- engage with open-science communities aligned with NASA's OSSI,
- coordinate with researchers and developers interested in regime-aware sensing,
- provide clear onboarding materials for external contributors.

All collaboration will follow open-source best practices, including transparent issue tracking, version control, and public documentation.

4. Project Timeline and Milestones

The 12-month performance period is divided into four phases:

Months 1–3: Protocol Finalization

- finalize dsrsp/0.1
- complete RSM and vST alignment layer
- publish schemas and examples

Months 3–6: Reference Implementations

- implement dsrsp/0.1 libraries
- implement RSM and vST modules
- develop ILP module

Months 6–9: Integration and Testing

- integrate dsrsp/0.1 into simulation environments
- validate classification and validation workflows
- produce example datasets and notebooks

Months 9–12: Documentation and Release

- publish developer documentation
- release integration guides
- host virtual workshop
- archive all deliverables with DOIs

This timeline ensures steady progress, clear checkpoints, and a well-defined public release.

5. Risk Management

The project includes the following risk mitigation strategies:

Technical Risks

- *Risk:* complexity of integrating multiple components (dsrsp, RSM, vST).
Mitigation: modular design, incremental releases, early testing.
- *Risk:* variability in sensor or simulation environments.
Mitigation: provide synthetic datasets and reference examples.

Schedule Risks

- *Risk:* delays in documentation or integration.
Mitigation: parallel workstreams and early drafting of documentation.

Sustainability Risks

- *Risk:* long-term maintenance beyond the performance period.
Mitigation: permissive licensing, community onboarding, and public repositories.
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6. Facilities, Resources, and Tools

The project requires:

- standard development workstations,
- open-source software tools,
- cloud-based repository hosting (GitHub),
- archival services (Zenodo),
- simulation environments for testing (open-source engines).

No specialized hardware or laboratory facilities are required.

7. Summary

The TriadicFrameworks project is structured to deliver a coherent, openly licensed scientific substrate through a clear, feasible, and well-managed plan. The PI will oversee all technical and organizational aspects, supported by modular workstreams, transparent development practices, and community engagement.

This management plan ensures that all HPOSS deliverables will be completed on schedule, openly accessible, and aligned with NASA's open-science mission.